Chronic Diarrhea

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Overview

- Introduction
- Approach to chronic diarrhea

- *Irritable bowel syndrome*
- *Inflammatory bowel disease*

Definition of diarrhea

- Bowel habits vary widely between individuals
- Must first evaluate patient's normal bowel patterns and nature of symptoms
- Diarrhea:
  - increase in daily stool volume, freq and fluidity
  - Stool weight > 250g/24h
  - >2-3 times/day or liquidity

Diarrhea

Acute

- Inflammatory
- Non-inflammatory

Chronic

- Osmotic
- Malabsorptive
- Secretory
- Inflammatory
- Motility
- Infective

Chronic diarrhea

- loose stools that last for >4 weeks
- usually means ≥3 loose stools/day

Chronic Diarrhea

- Osmotic diarrhea
  - Causes
    - Lactase deficiency
    - Laxative abuse
    - Malabsorption
  - Resolves with fasting
- Malabsorption
  - Causes: small bowel diseases, resection, bacterial overgrowth and pancreatic diseases
  - Weight loss, osmotic diarrhea and nutrient deficiency
- Secretory
  - Causes: endocrine tumors, bile salt malabsorption and laxative abuse
  - Large volume watery diarrhea
  - Persists in fasting state
- Inflammatory
  - Causes: Crohn's, UC
- Motility
  - Irritable bowel syndrome
- Chronic infection

Mechanisms of Diarrhea

- Secretory
  - persists despite fasting
  - no pus, blood or excess fat
- Exudative or inflammatory
  - mucus and blood
  - PMN in stool
- Decreased absorption
  - osmotic (ceases after fasting)
  - decreased in absorption surface motility disorder
  - motility disorder

Drug induced diarrhea

- Antacid, acid suppressing (Mg-containing, H2-RA, PPI)
- Alcohol
- Antibiotic
- Antihypertensive (beta blockers)
- Anti-inflammatory (NSAID, 5-aminosalicylates)
- Colchicine, misoprostol, theophylline
- Vitamin and mineral supplements, herbal products
- Coffee, tea or colas
  (caffeine or methyxanthine-induced diarrhea)
- Dietetic foods, gums or mints
  (sorbitol and mannitol related osmotic diarrhea)

Physical examination

- Extent of fluid and electrolyte depletion
- Nutritional status
- Vitamin def
- Causes of diarrhea
  - skin rash and flushing
  - thyroid masses, toxic signs
  - mouth ulcers
  - arthritis
  - hepatomegaly
  - anorectal exam - mass, anal tone, perianal diseases
- Signs of toxicity
  - fever
  - distended, rigid or tender abdomen

Specific Testing

- Directed by history and physical exam
- Standard investigations
  - Blood tests
  - Stool tests
  - Imaging
  - Endoscopy and biopsy

Laboratory tests

- Complete blood count
  - Anemia (GI blood loss)
  - Leucocytosis (inflammation)
  - Eosinophilia (neoplasms, allergy, collagen vascular diseases, parasite, eosinophilic gastroenteritis)
- Inflammatory markers
  - ESR, CRP
- Serum electrolytes, liver function tests
  - Na, K, Ca, P, Mg, Albumin, glucose
- Others
  - ANF, p-ANCA, serum Ig levels
  - HIV
  - TSH

Stool analysis (spot)

- Stool for culture & micro

  Cl. difficile toxin

  Aeromonas, Plesiomonas; protozoan and

  parasitic infection;

  Giardia
- Stool for leucocytes
  - inflammatory cause
- Stool for occult blood
- Fecal calprotectin

- Stool for fat
  - malabsorption or maldigestion
- Stool for Na
  - osmotic diarrhea (low Na <60 mmol/L)
  - secretory diarrhea (high Na >90 mmol/L)
- Stool for pH
  - <5.6 carbohydrate malabsorption
  - <5.3, Osmotic diarrhea

Fecal Calprotectin

- 24kDa dimer of Ca binding proteins
- Indicates migration of neutrophils to the intestinal mucosa
- Elevated in
  - Infectious diarrhea
  - Crohn's disease
  - UC
  - Cancer
- Stable in room temperature

Protein losing enteropathy

e.g., IBD, sprue, Whipple's disease, allergic gastroenteropathy, intestinal lymphangiectasia, constrictive pericarditis, congenital hypogammaglobulinemia, etc

<u>Diagnostic Workup</u>:

- Labeled human serum scan
  - Localization of source
- <u>Fecal alpha-1-antitrypsin concentration</u>
  - Suggestive of excessive GI protein loss
- <u>Serum alpha-1-AT clearance</u>

Radiological imaging

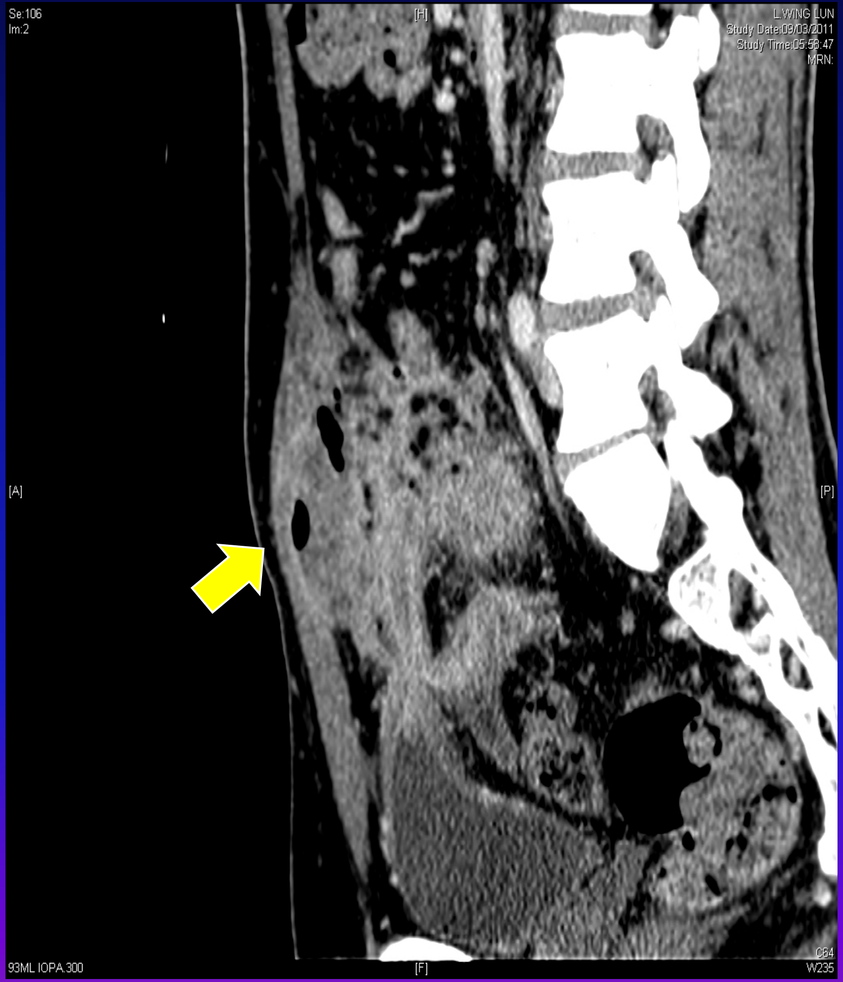
	Plain X-ray	calcification chronic pancreatitis
Barium studies	Barium meal	non-specific diagnosis
	Small bowel follow through	small bowel and ileal abnormalities in Crohn's disease
	Barium enema	colon cancer and polyp, mucosal diseases like IBD, ischaemia and radiation, laxative abuse
	Ultrasound	biliary tract obstruction pancreatic disease
	CT /MRI (Enterography)	IBD complications
	Lymphagiogram	Lymphagiectasia

Small bowel (Ba) follow through

Long segments of stricture in patients with Crohn's Disease

SBFT vs CT


L
PRONE
95 MINS.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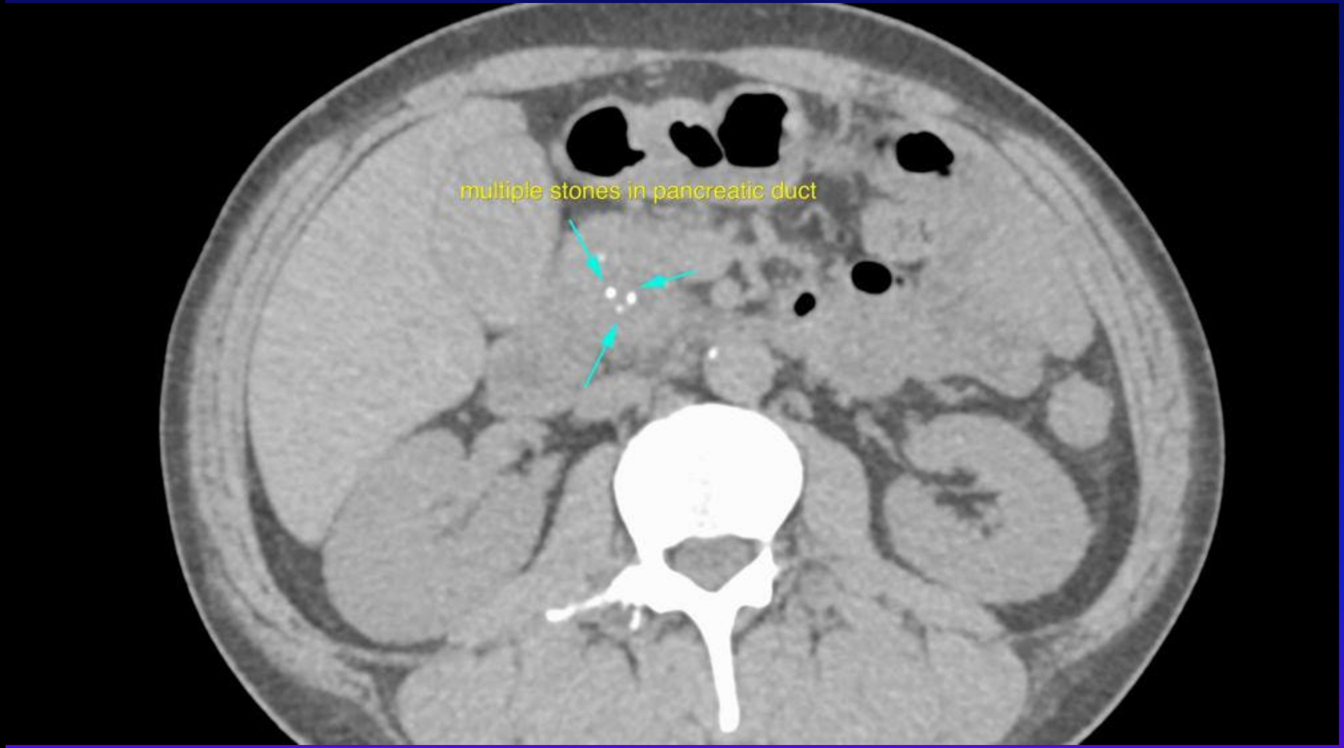
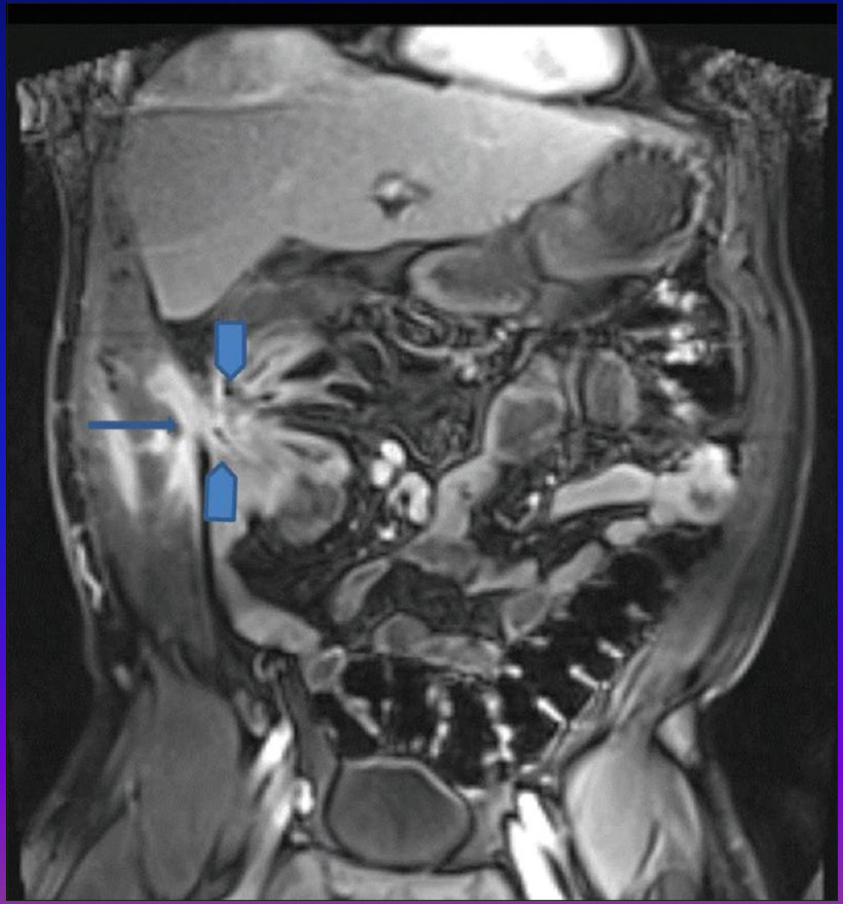
Chronic pancreatitis



MR Enterography

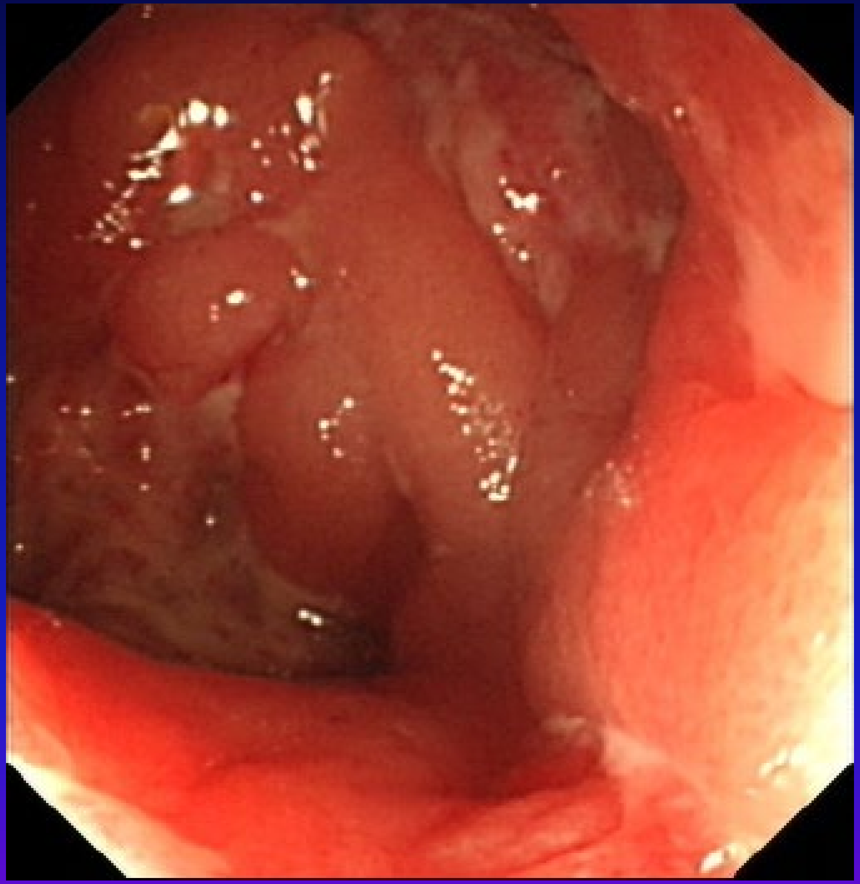
- Assessing SB mucosal inflammation, fistula, abscess and stricture
- Radiation free
- Need oral contrast to distend SB



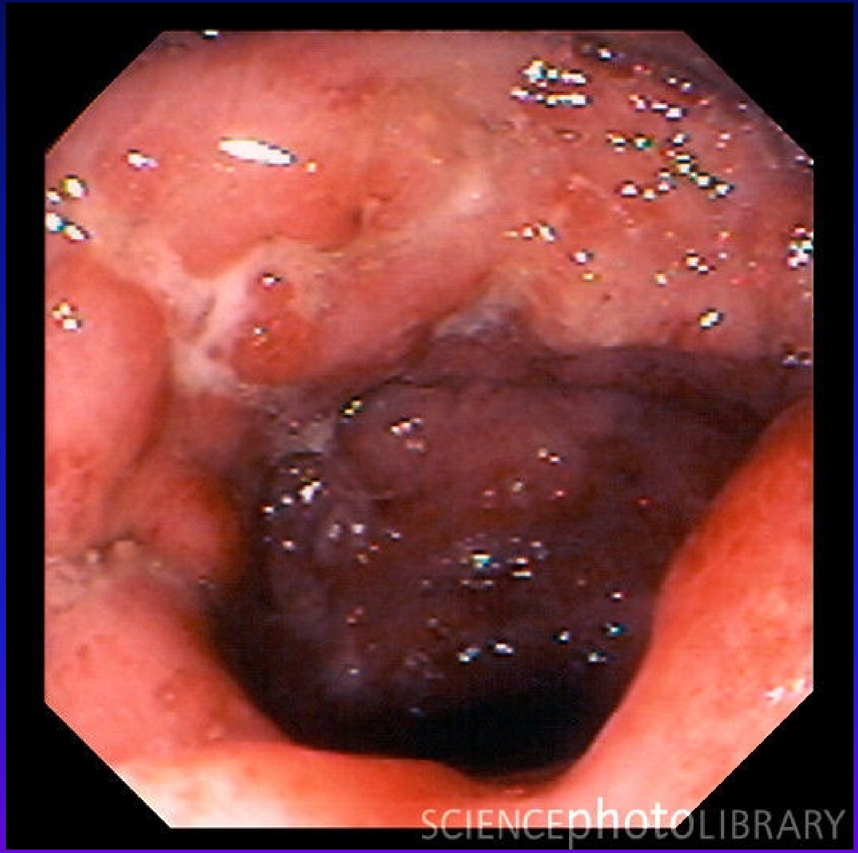
Endoscopy

- Upper: OGD
  - Duodenal biopsy (Celiac, Whipple, Crohn's)
- Lower: Sigmoidoscopy and Colonoscopy
  - Obtaining mucosal biopsy
    - IBD
    - Opportunistic infections
    - Microscopic colitis
- Mid: Small bowel enteroscopy / capsule endoscopy

Inflammatory Bowel Disease



Colonoscopy in IBD

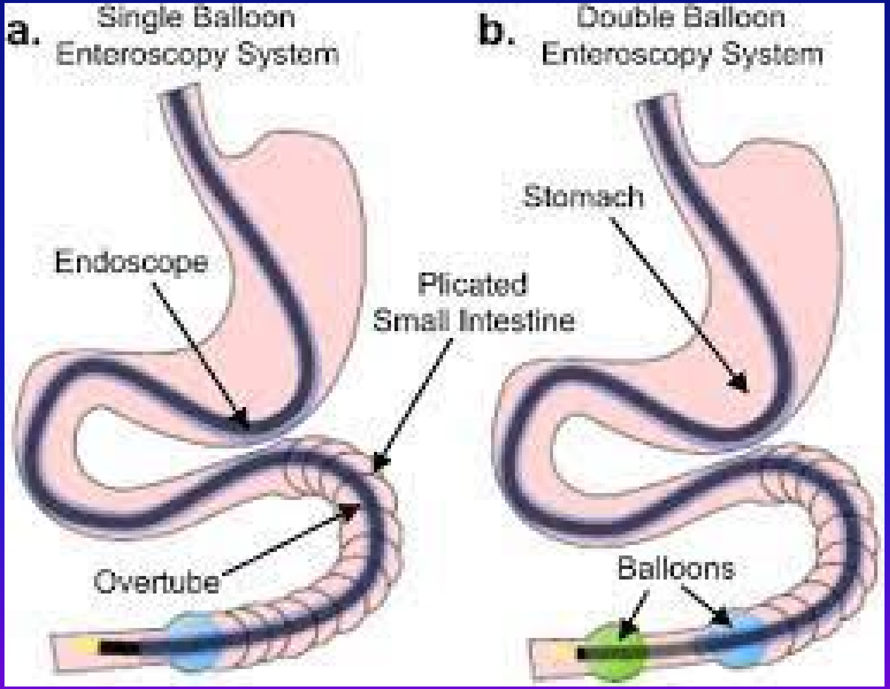


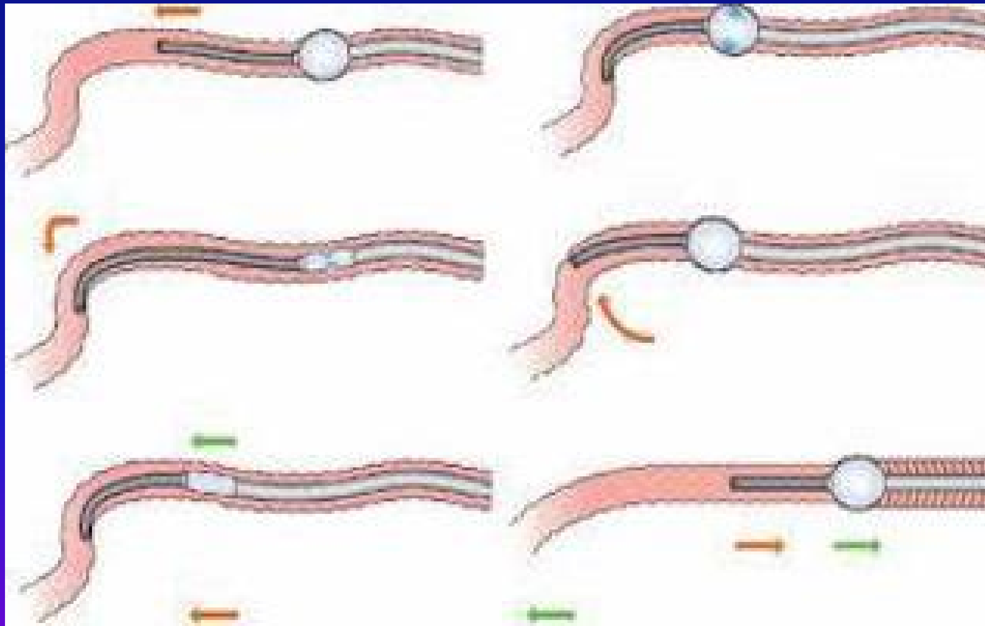
Crohn's colitis

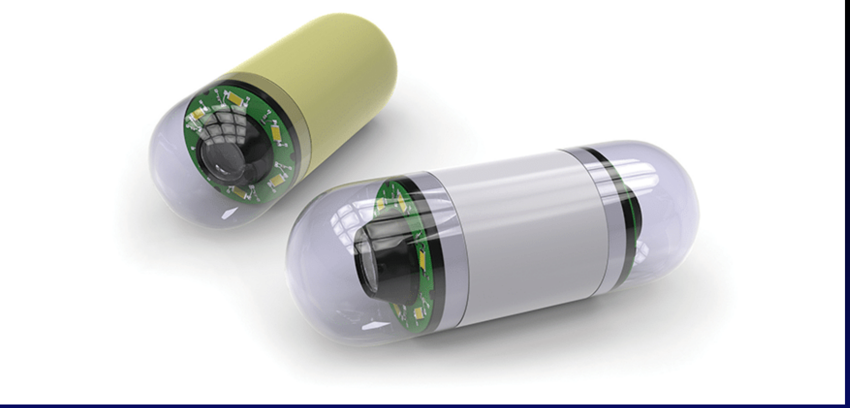
Crohn's ileitis

Enteroscopy

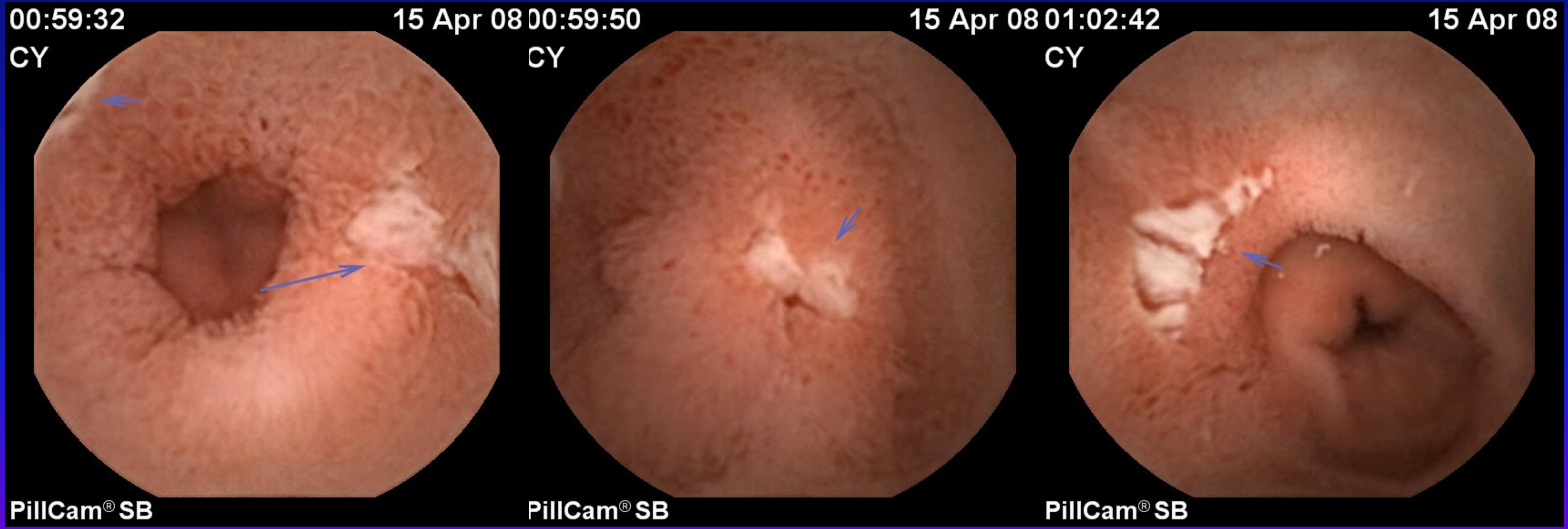
- Balloon enteroscopy







Capsule endoscopy in CD



Management of chronic diarrhea

- Specific treatment
  - *direct to the cause of the disease*

- Supportive treatment

? Antibiotics	*Anti-diarrheal drugs*
Octreotide	*Intraluminal absorbants - charcoal*
Bile acid binding resin	*Bismuth compounds*

Management of malabsorption

- Dietary supplements
  - Ca, Mg, K, Fe, Folate,
  - Vit A, D, K, B12
- Anti-diarrheal agents
  - Cholestyramine (bile salt related diarrhea)
  - Lomotil, Loperamide
- Pancreatic enzymes supplement
- Enteral and parenteral supplementation

Irritable bowel syndrome

Irritable Bowel Syndrome

Abdominal Pain

- **relieved by defecation**
- **altered frequency**
- **altered consistency**

Rome IV criteria*

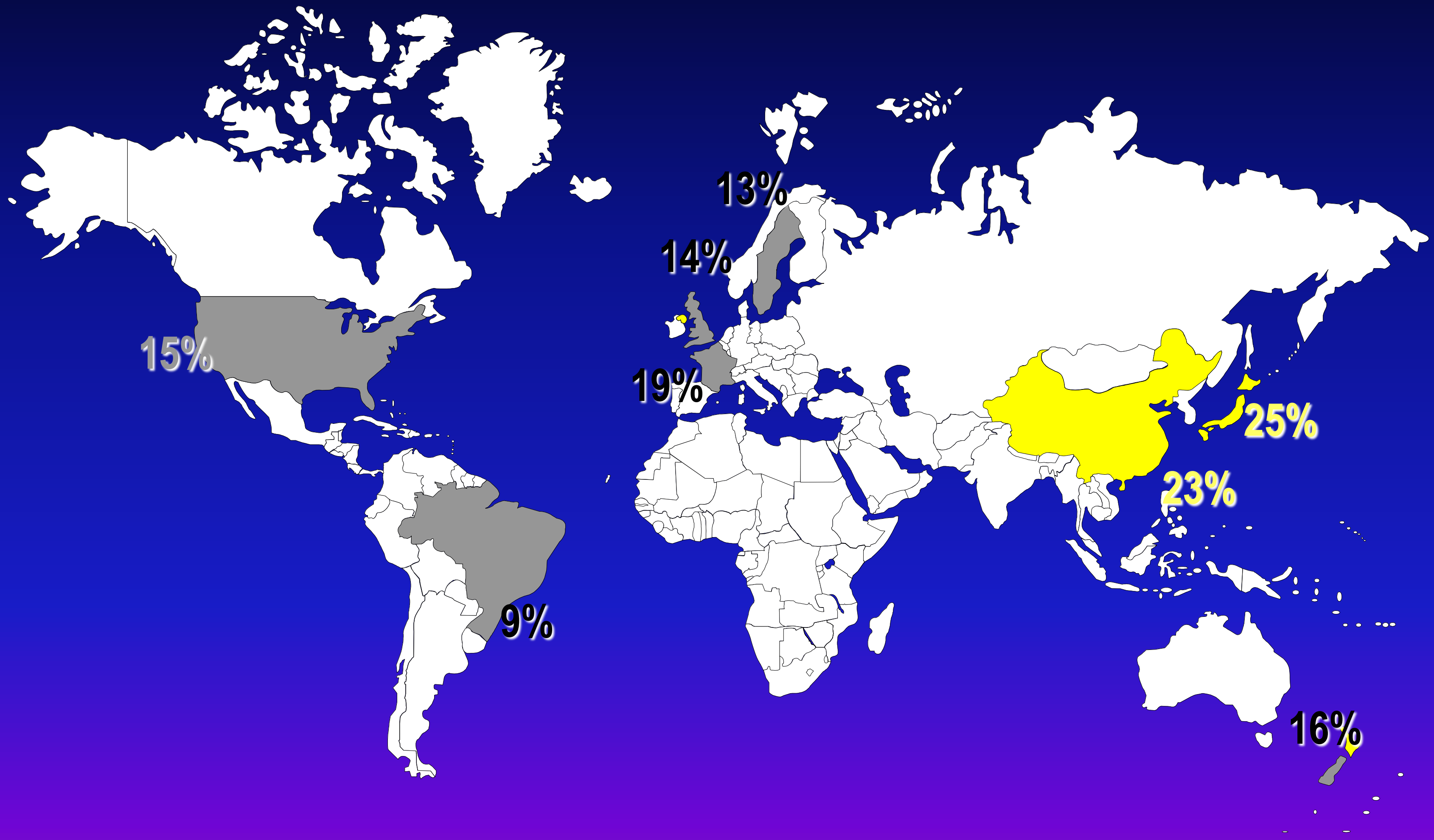
Recurrent abd pain, on average, at least 1 day/week in the last 3 months, associated with ≥ 2 of the following:

1. Related to defecation
2. Associated with a change in frequency of stool
3. Associated with a change in form (appearance) of stool

*For the last 3 months with symptom onset at least 6 months before diagnosis

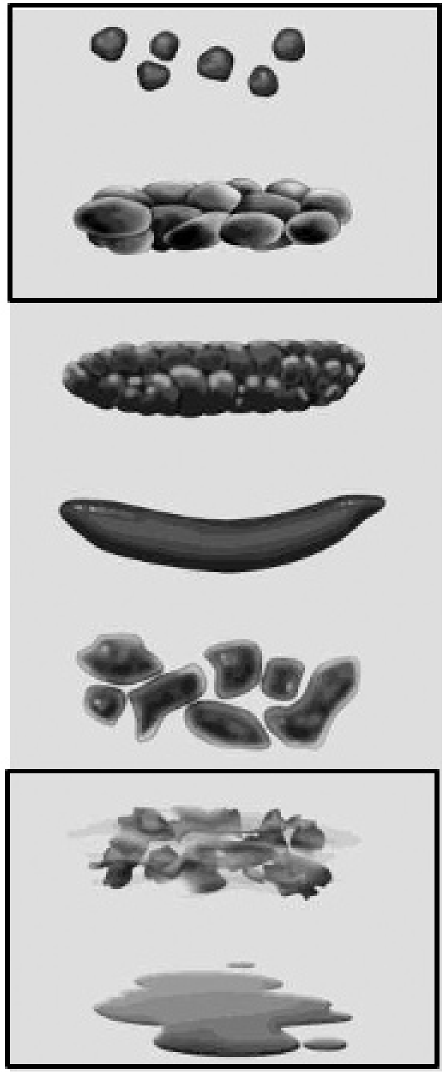
Lacy BE *et al*, Gastroenterol 2016

IBS - Population prevalence



Pathophysiological features

- Brain Gut axis
- Autonomic nervous system
- Altered bowel motility [motor]
- Visceral hypersensitivity [sensory]
- Psychosocial factors
- Neurotransmitter imbalance (serotonin)



Type 1	Separate hard lumps, like nuts (hard to pass)
Type 2	Sausage-shaped but lumpy
Type 3	Like a sausage but with cracks on the surface
Type 4	Like a sausage or snake, smooth and soft
Type 5	Soft blobs with clear-cut edges
Type 6	Fluffy pieces with ragged edges, a mushy stool
Type 7	Watery, no solid pieces, entirely liquid

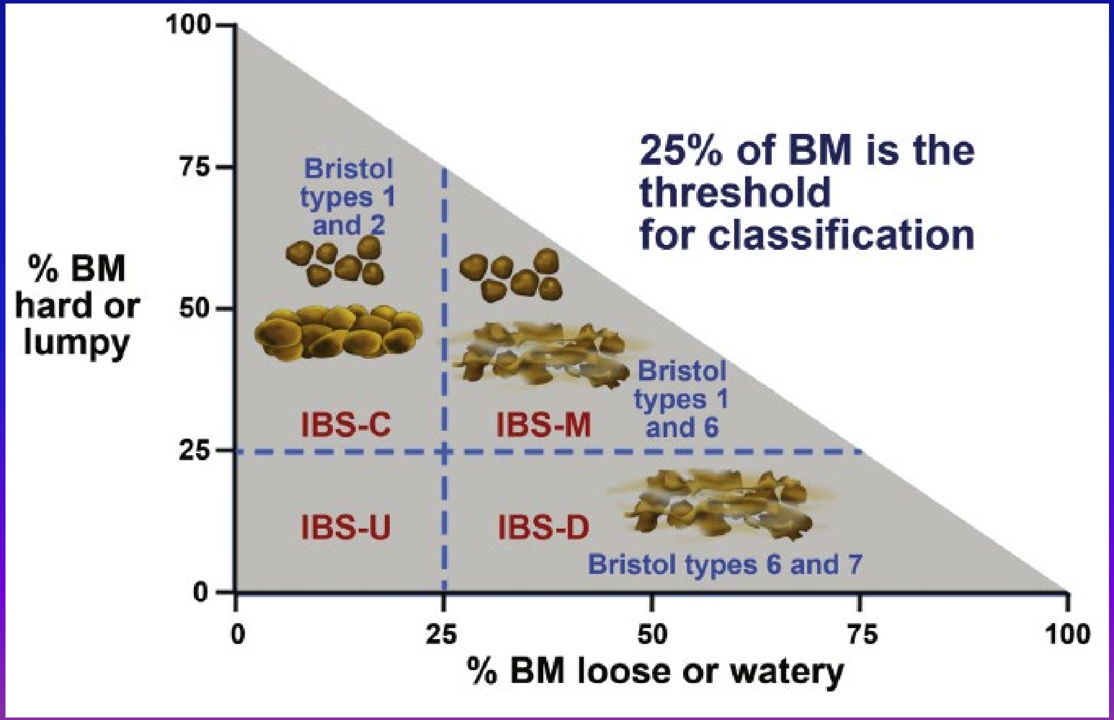
IBS Subtypes

IBS with diarrhea [IBS-D]: loose or watery stool >=25% and hard stool <25% of bowel movements

IBS with constipation [IBS-C]: hard stool >=25% and loose or watery stool <25% of bowel movements

Mixed IBS [IBS-M]: hard stool >=25% and loose or watery stool >=25% of bowel movements

Unsubtyped IBS: insufficient abnormality of stool consistency to meet criteria for IBS-C, D or M.



Features Against Irritable Bowel Syndrome

HISTORY

- Weight loss
- Rectal bleeding
- Onset in older patients
- Family history of CA colon or IBD

INVESTIGATIONS

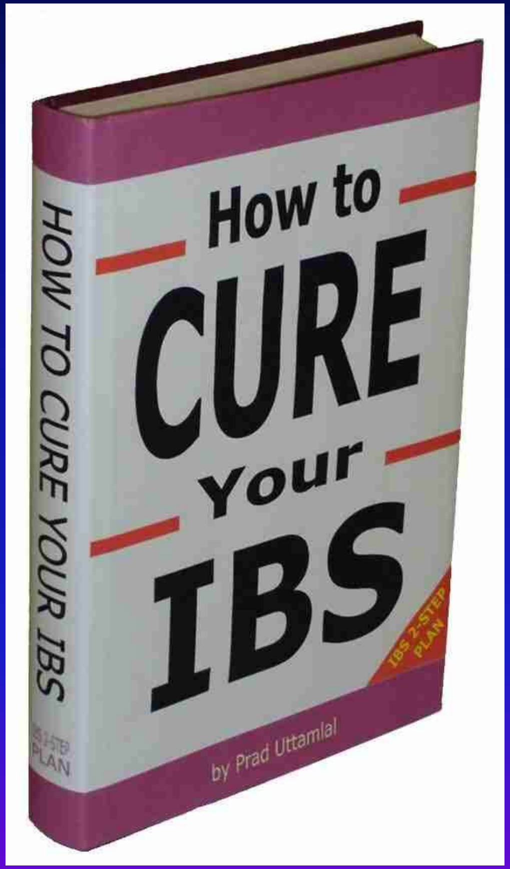
- Positive occult stool
- Anemia
- ↑ WBC
- ↑ ESR
- Abnormal biochemistry

Management of IBS

- Multi-faceted approach
  - Education
  - Reassurance
  - Dietary modifications
  - Pharmacotherapy
  - Psychological treatments

Medical Treatment of IBS

- Targets predominant symptoms of IBS
  - Abdominal pain
  - Diarrhea
  - Constipation



Treatment (symptomatic)

Diarrhea

1. Opiod agonists (loperamide)
2. Diet (Low FODMAP)
3. Bile salt sequestrants (cholestyramine)
4. Probiotics
5. Antibiotics (Rifaximin)

Constipation

1. Psyllium
2. PEG
3. Prucalopride (5HT4 agonist)
4. Chloride channel activator (lubiprostone)
5. Guanylate cyclase C agonists (linaclotide)

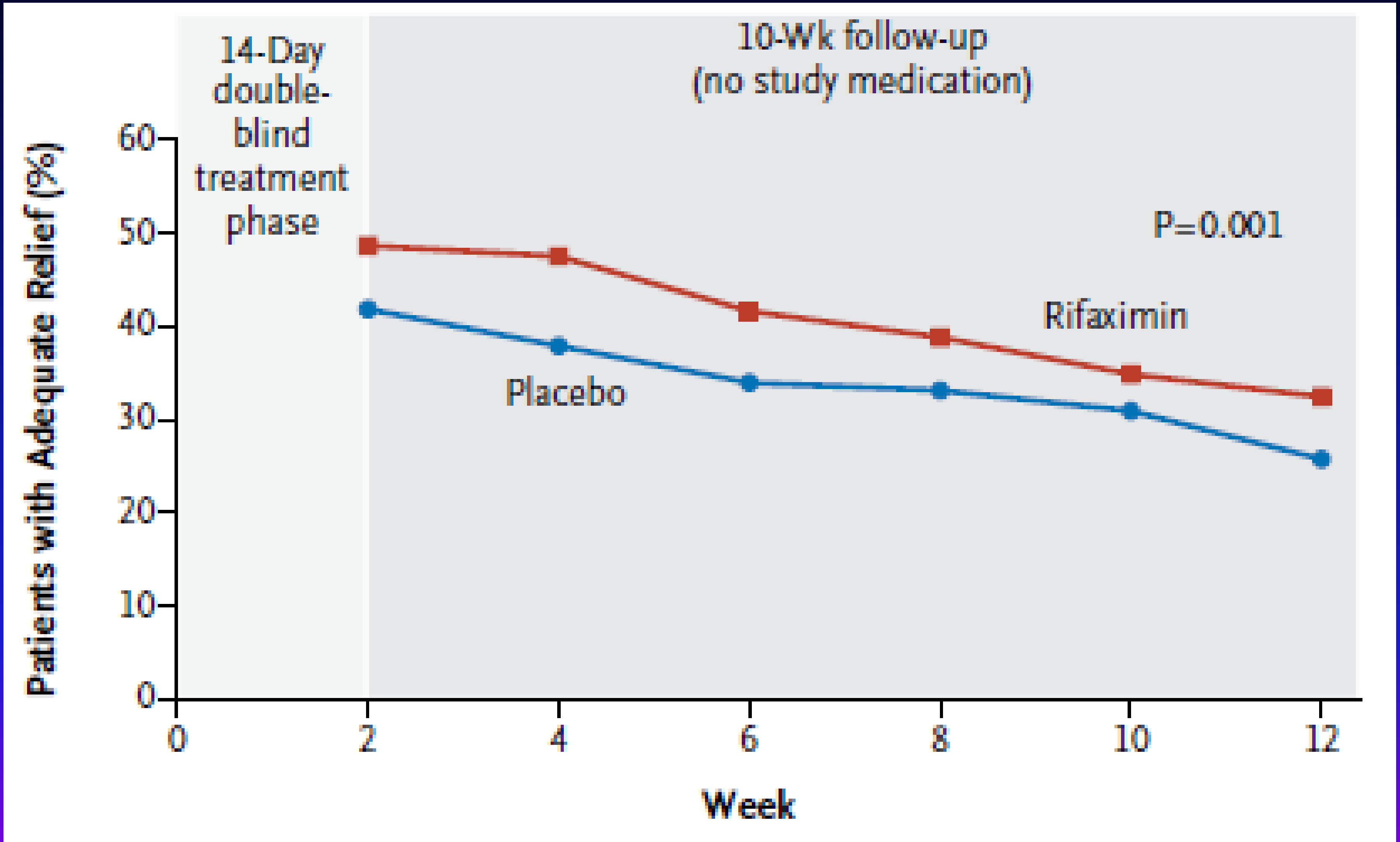
Treatment (symptomatic)

Pain

1. Smooth muscle antispasmodics: otilonium, mebeverine
2. Peppermint oil
3. TCA: amitriptyline, desipramine
4. SSRI: citalopram, paroxetine, sertraline
5. Chloride channel activator (lubiprostone)
6. Guanylate cyclase C agonists (linaclotide)

Antibiotics for IBS?

- Rifaxamin
  - a oral, nonsystemic, broad-spectrum antibiotic that targets the gut and is associated with a low risk of bacterial resistance



Piemental M et al. N Engl J Med 2011

Inflammatory bowel diseases

Inflammatory Bowel Disease (IBD)

Ulcerative colitis (UC)

- confined to mucosa & submucosa of colon, continuous lesion

Crohn's disease (CD)

-extends from mucosa to the serosa
-can involve the whole GI tract
-skipped lesions

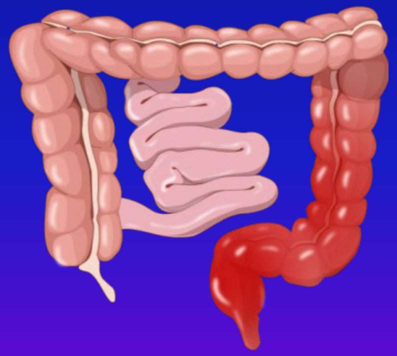
Indeterminate colitis

Inflammatory Bowel Disease (IBD)

Inflammatory Condition of the Intestine[1]

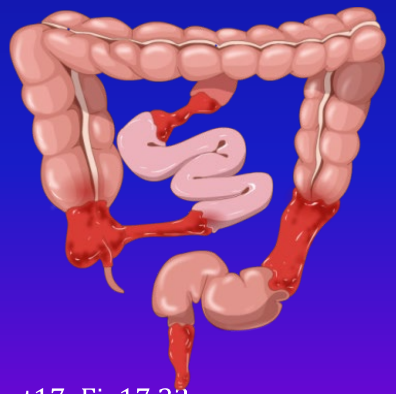
Ulcerative colitis (UC)

Usually affects the rectum and the colon[1,2]



Crohn's disease (CD)

Affects any part of gastrointestinal tract[1,2]

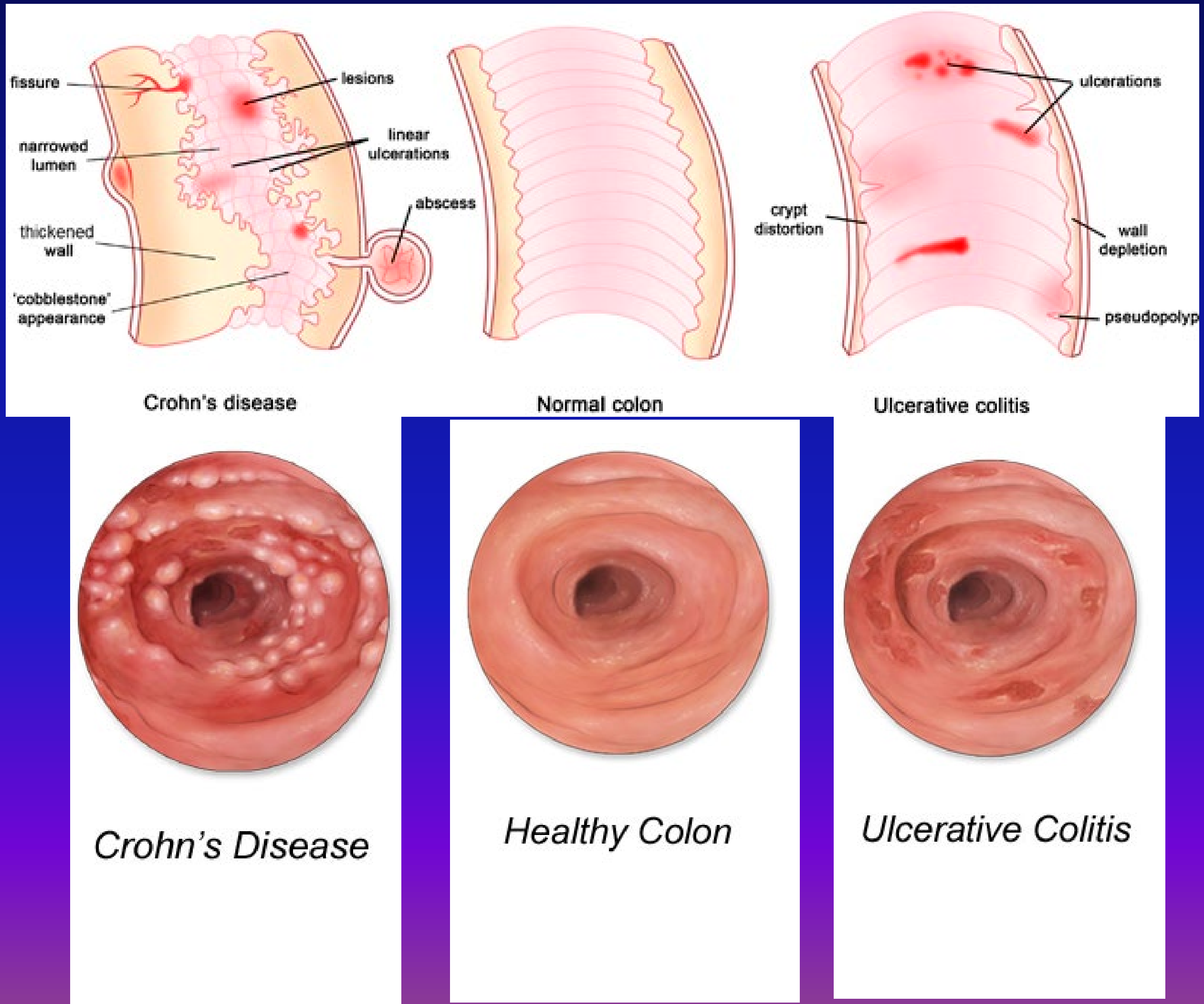


Adapted from *Robbins and Cotran Pathologic Basis of Disease*, 8th ed. Chapt17; Fig17.32

GI, gastrointestinal.

1. McQuaid, KR. *Current Medical Diagnosis & Treatment*. 52nd ed. New York: McGraw-Hill; 2013.
2. Friedman S et al. *Harrison's Principles of Internal Medicine*. 18th ed. New York: McGraw-Hill; 2012.

Inflammatory Bowel Disease (IBD)

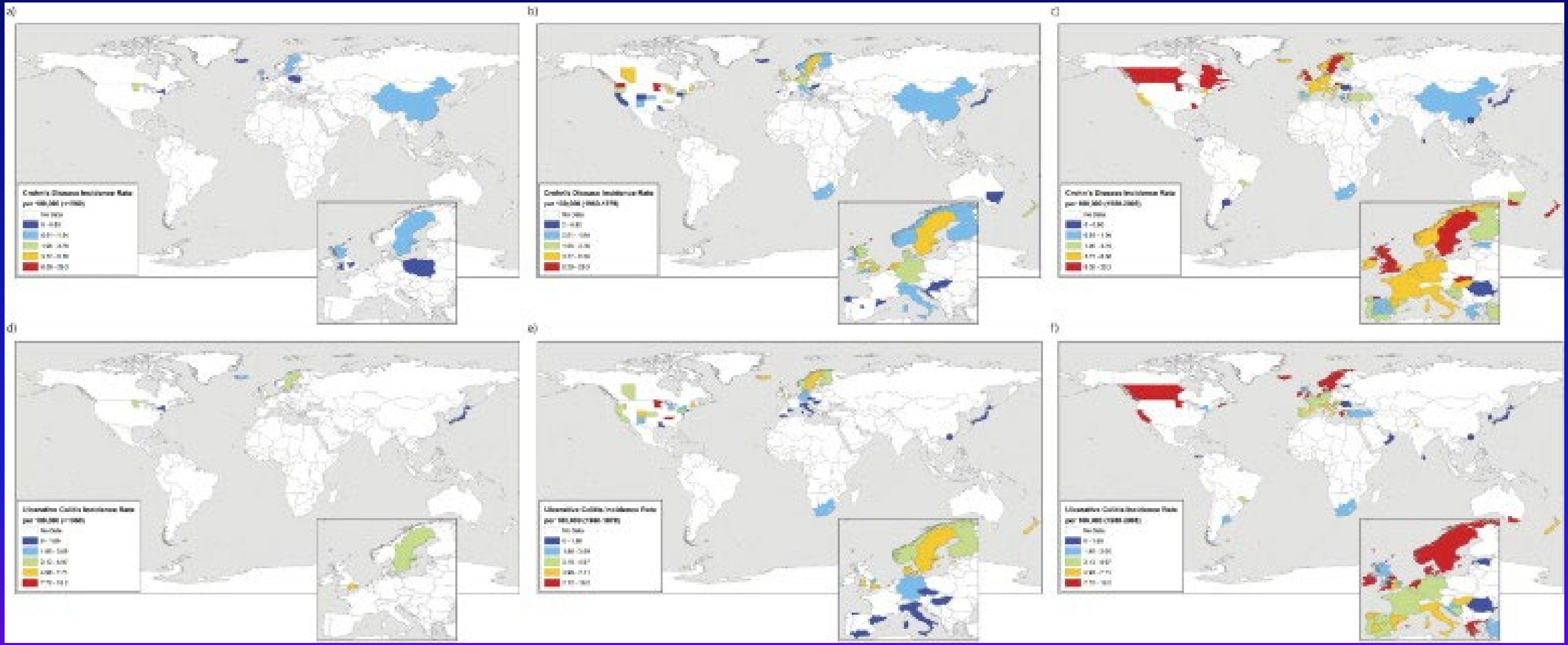


	UC	CD
Radiological	Diffuse and **continuous**	Patchy
	Rectosigmoid always involved	Rectosigmoid involves in 50%
	No small bowel involvement	Usually involves small bowel and terminal ileum
	No skip lesion No stricture and fistula	**Skip lesion** **Stricture and fistula**
Endoscopic	Hyperemic mucosa	Aphthous lesions
	Shallow ulcers Diffusely granular appearance	**Solitary and deep ulcers** Cobberstone appearance
Histology	Superficial inflammation	**Transmural** inflammation
	Continuous involvement	Patchy involvement
	Granuloma rarely seen	**Granuloma** common
	Globet cells depletion	Globet cells present

Increasing worldwide IBD incidence

CD

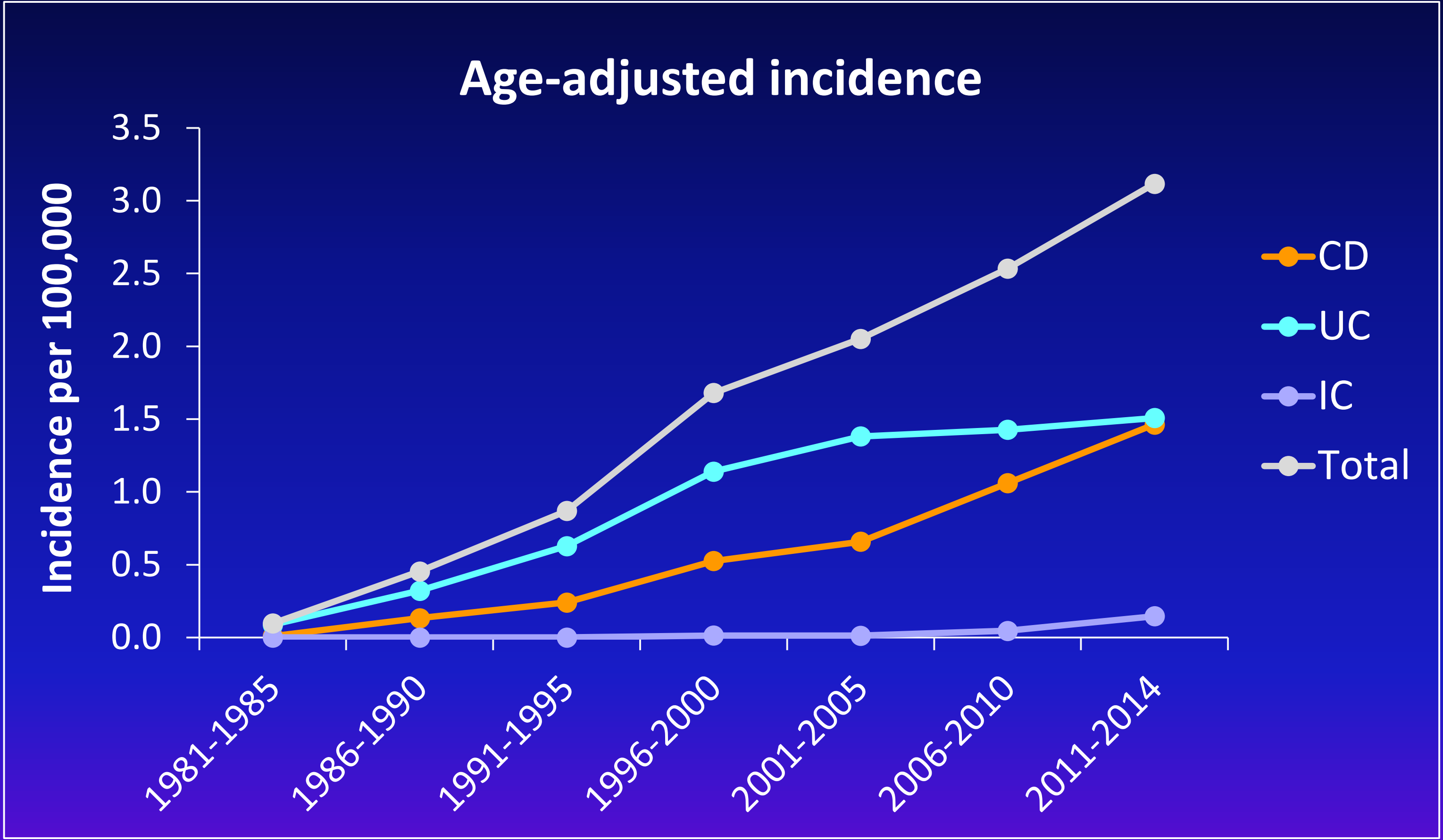
Before 1960 | 1960 – 1979 | fter 1980



UC

Molodecky NA, et al. Gastroenterol 2012

Hong Kong IBD incidence



Ng SC, Leung WK et al. IBD 2016

Hong Kong IBD patient characteristics

	Total N = 2575	CD N = 983	UC N = 1541	Indeterminate N = 51
Age (median, IQR)	49 (36–60)	40 (30–53)	52 (42–63)	50 (33–62)
Age of disease onset (median, IQR)	37 (27–50)	30 (22–43)	41 (31–52)	46 (29–55)
Male	59.4%	65.0%	56.1%	51.0%
Duration of disease since diagnosis (median, IQR/months)	101 (45–173)	84 (39–158)	117 (55–184)	38 (16–72)
Family history	3.0%	2.9%	3.1%	0
Hb (median, IQR (g/dL))	12.4 (10.6–13.7)	11.6 (10.1–13.0)	12.9 (11.3–14.1)	13.1 (11.2–14.4)

Ng SC, Leung WK et al. IBD 2016

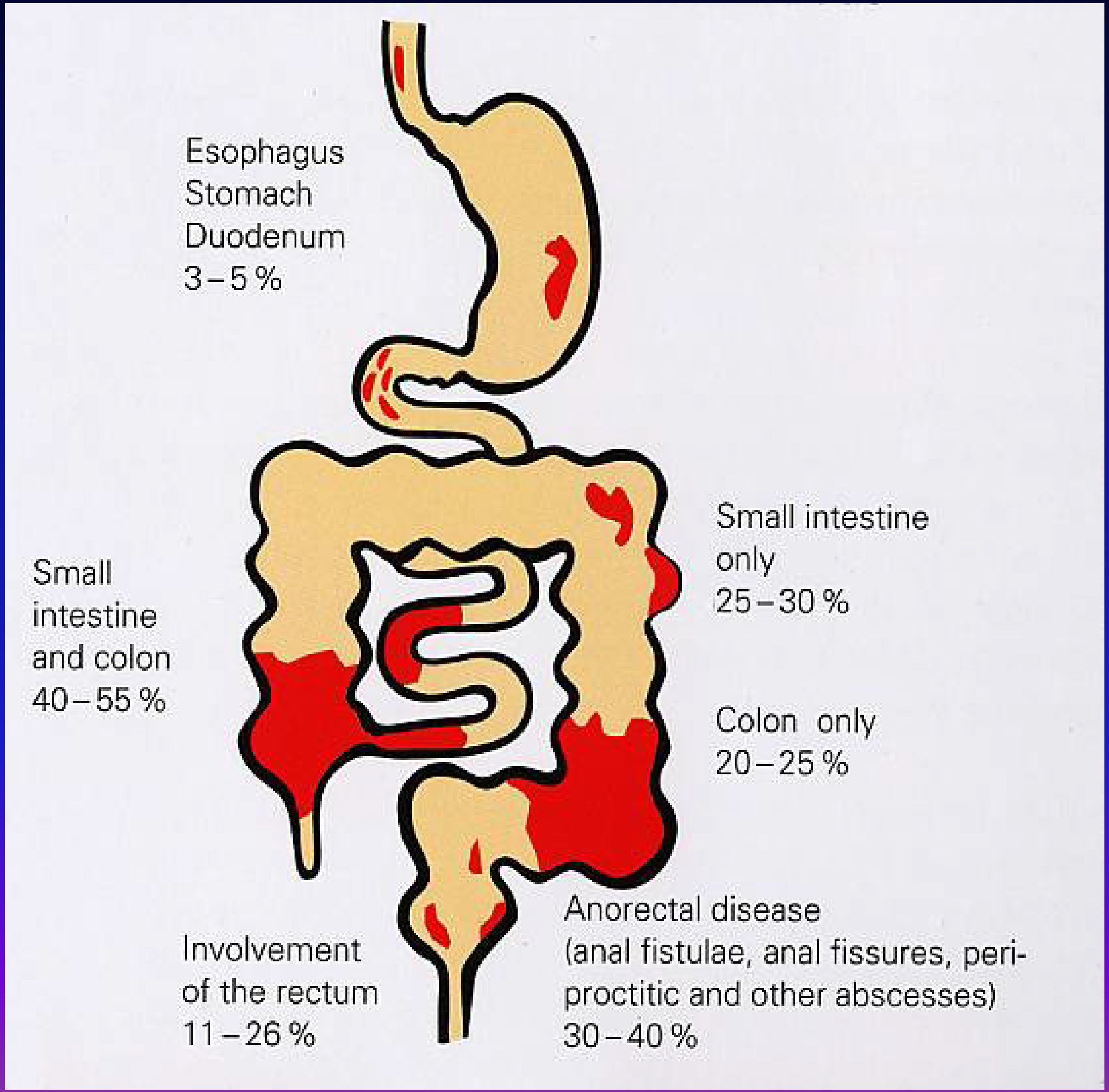
Etiology

- Generally unknown
- Family aggregation of both UC and CD, suggesting genetic basis in both diseases and partially sharing the genetic basis
- Environmental (diet, infection)
- Immunological
- Gut microbiota

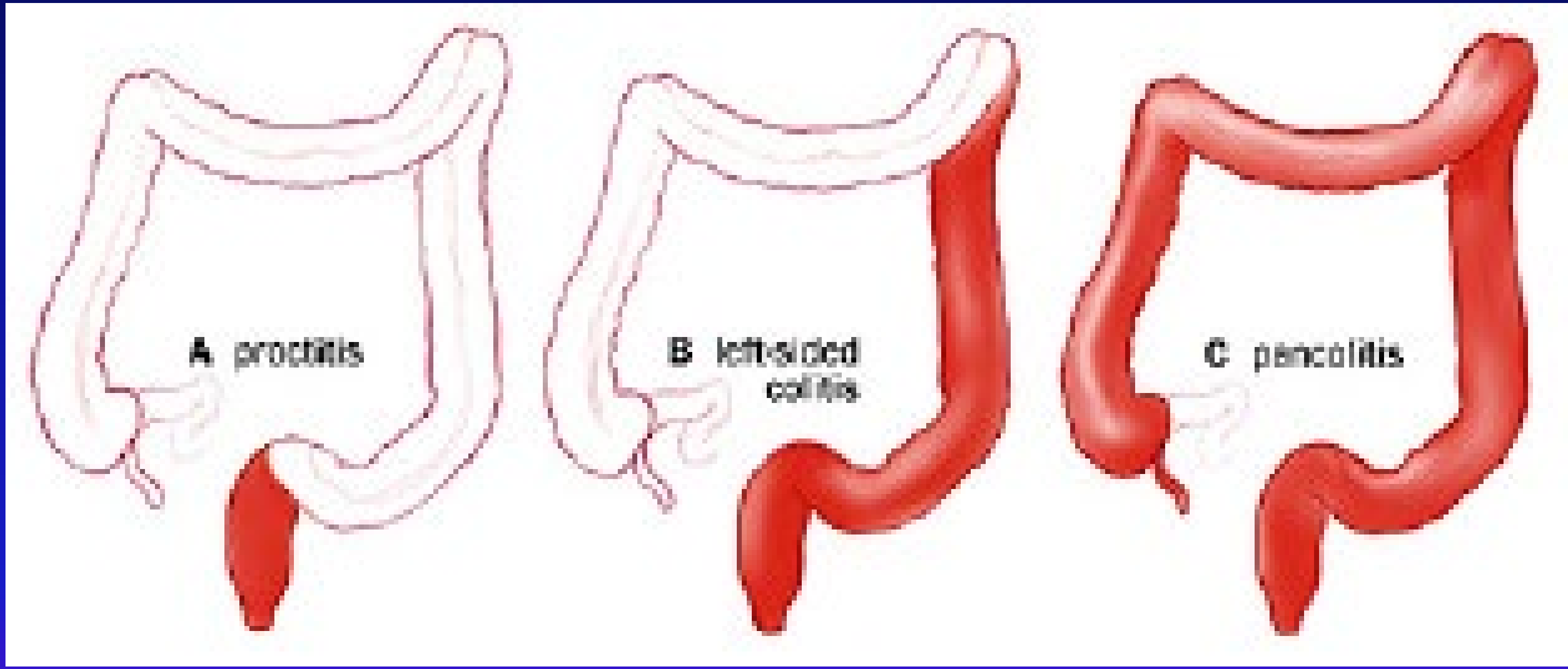
Environmental & Genetic Risk factors

Risk Factors	UC	CD
Environmental		
▪ smoking	↓	↑
▪ appendicectomy	↓	?no effect
Genetic		
▪ MHC genes	Complicated	Complicated
▪ NOD2 gene SNP8, 12, 13	no effect	↑
▪ other genes by linkage analysis	Complicated	Complicated

Anatomical location of Crohn's disease



UC: Disease Phenotype



Proctitis

Left-sided colitis

Extensive colitis

Vienna and Montreal Classification of CD

	Vienna (1998)	Montreal (2005)
Location	L1 ileal	L1 ileal
	L2 colonic	L2 colonic
	L3 ileocolonic	L3 ileocolonic
	L4 upper	L4 isolated upper disease*
Behaviour	B1 non-stricturing, non-penetrating	B1 non-stricturing, non-penetrating
	B2 stricturing	B2 stricturing
	B3 penetrating	B3 penetrating
		p perianal disease modifier†

Satsangi J et al. Gut 2006

Extraintestinal complications

- **Arthritis, peripheral arthropathy, spondylitis like ankylosing spondylitis**
- **Biliary tract complications**
  - Gallstones, primary sclerosing cholangitis
- **Renal complications**
  - Kidney stones
- **Skin complications**
  - Pyoderma gangrenosum
  - Erythema nodosunm
  - Aphthous ulcers of mouth
- **Eye complications**
  - Uveitis, iritis, episcleritis, scleritis

Systemic Complications In Ulcerative Colitis

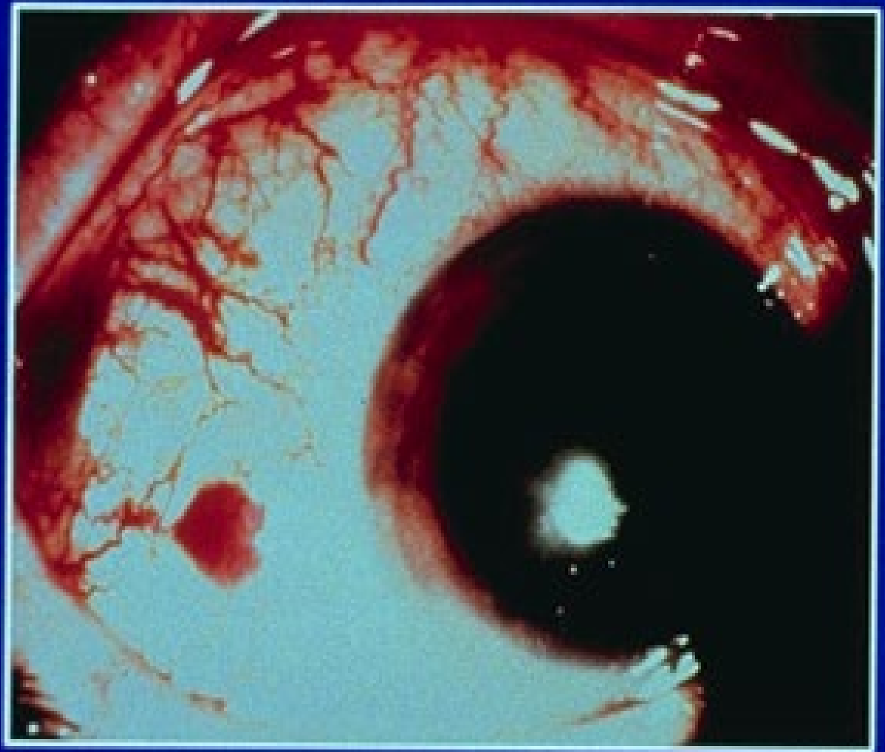
SKIN

Erythema Nodosum

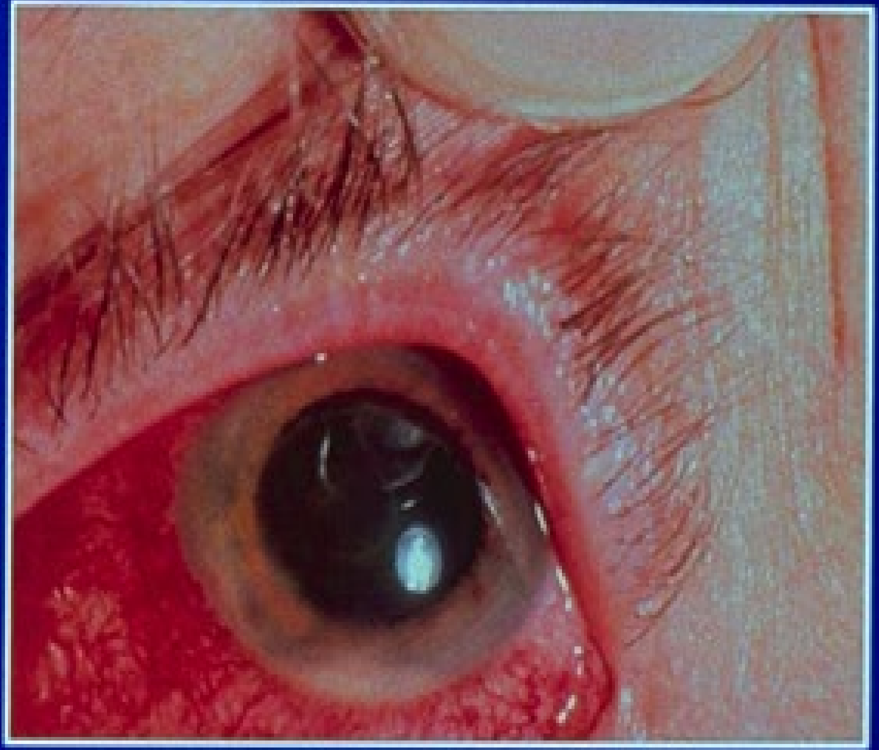
Pyoderma Gangrenosum


Systemic Complications Of Ulcerative Colitis

EYE



Episcleritis



Uveitis

Complications of CD

- **Malnutrition**

  Protein, calorie and vitamin deficiencies

  Poor intake, protein losing enteropathy, malabsorption
- **Abscesses and fistula**

  extension of the mucosal fissure and ulcer through the bowel wall into the extra-intestinal tissue. Abscess: peritoneal cavity; Fistula: adjacent viscera, bladder, vagina and abdominal wall.
- **Stricture / obstruction**

  mucosal thickening due to active inflammation, scarring, adhesions, food impaction in a long-standing stricture
- **Perianal disease**

  Perianal abscess or fistula, anal fissures

Complications of UC

Toxic megacolon

- Diagnosed by plain AXR
- Clinical features of severe UC: fever > 38, heart rate > 120/min, Anaemia, Low albumin.
- Exclude coexisting C. difficile or CMV infection
- Treatment: bowel rest and TPN, fluid and electrolyte replacement, corticoteroids, close monitoring.
- Outcome: around 50% responded to medical tx, around 50% go to urgent colectomy if not responding in 3-7 days.

Colorectal cancer

AXR – Thumb printing

AXR – Toxic Megacolon

Malignancy

- Moderately increased risk of colon cancer in patients with ulcerative colitis
- Extensive disease: 8 – 10 years after onset
- Need regular endosocpic surveillance in long-standing extensive disease [every 1-2 years]
- Higher risk in patients with concurrent primary sclerosing cholangitis

Diagnosis of IBD

- No single test or gold standard
- Diagnosed by endoscopic; radiological and pathological and/or biochemical findings
- Rule out infection, esp TB

Investigations

- CBC – anaemia, high wbc
- CRP
- Stool – wbc, bacteria, parasites
- Serology
  - Anti-saccharomyces cerevisiae antibodies (ASCA)
  - pANCA
- Colonoscopy / sigmoidoscopy and bx
  - CD: Granuloma
  - UC: Mucosal inflammation
    - Chronic changes like glandular distortion and atrophy
- Small bowel series (CD)
- CT or MR enteroclysis/enterography (CD)
- CT scan – for diagnosis & complications

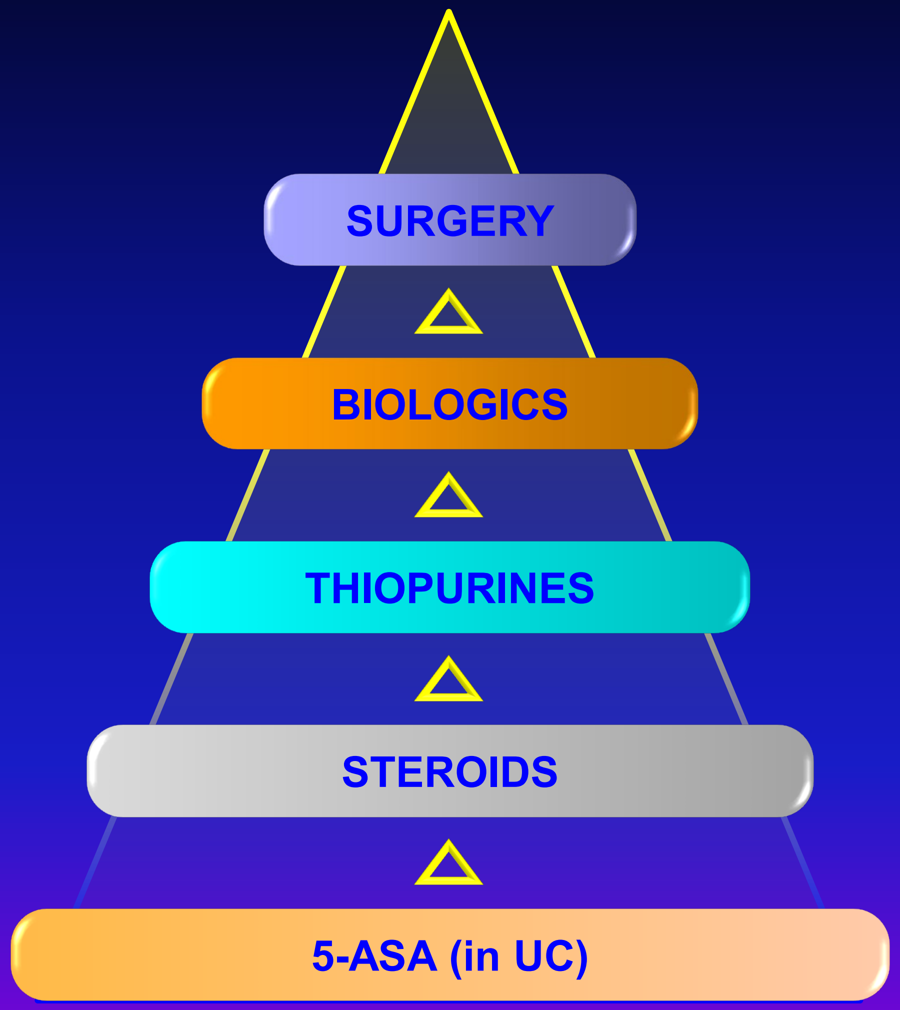
Treatment of IBD

- Induction of remission during acute flare up
- Maintenance of remission
- Modification of clinical course
  - Reduce surgery and complications
  - Improve nutritional status
  - Cancer prevention
  - Improve QoL

Medications

1. 5-aminosalicylic acid
2. Steroid
3. Immunomodulators
4. Biologics

Traditional approach to IBD management



Bouguen G, *et al.* Clin Gastroenterol Hepatol 2015;13:1042-50.

5- Amino-Salicylates

- Indications:
  - Mild to moderately severe UC and colonic CD
  - Maintenance of remission
- Local anti-inflammatory action
- Sulphasalazine
  - sulphapyridine + 5ASA
  - Side effects: skin rash, hemolysis, neutropenia, male infertility, pancreatitis
- 5 ASA analogues
  - Mesalazine, olsalazine
    - Oral, enema or suppository

Immunosupressive therapy: Azathioprine, 6-Mercaptopurine

- Indications:
  - Frequently relapsing disease
  - Steroid sparing effects
  - Fistulating CD
- delayed onset 3 months
- well tolerated, side effects 10%
  - Bone marrow suppression, allergy, hepatotoxicity, pancreatitis

Biologics for IBD

- Anti-TNFα
  - Infliximab
  - Adalimumab
  - Certolizumab Pegol
- Anti-adhesion molecules
  - Vedolizumab
- Anti-IL23
  - Ustekinumab
  - Risankizumab
- JAK inhibitor
  - Tofacitinib
  - Upadacitinib

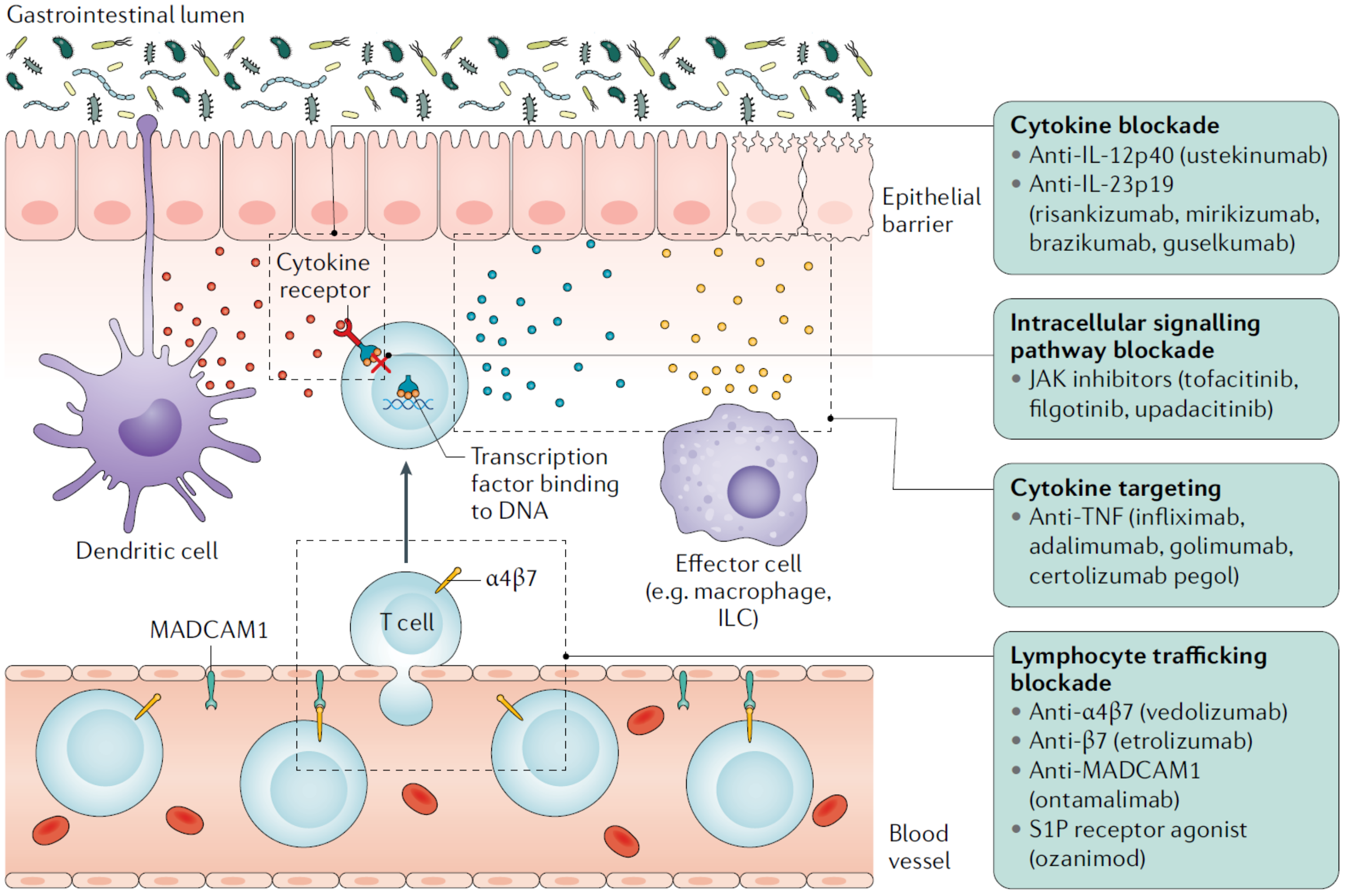
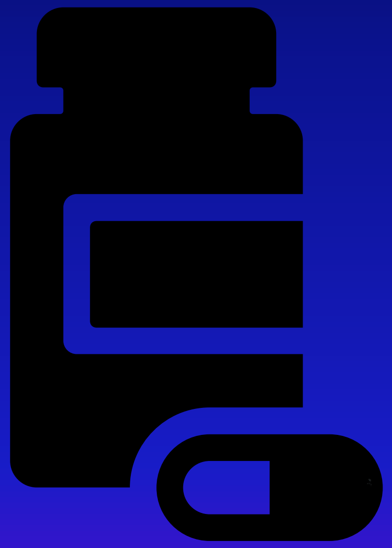


Fig. 1 | **Drug targets in IBD.** Drug targets in IBD can be broadly separated into: cytokine blockade (for example, IL-12p40 with ustekinumab, IL-23p19 with risankizumab, mirikizumab, guselkumab or brazikumab); T cell intracellular signalling pathway blockade (for example, JAK inhibitors such as tofacitinib, folgotinib or upadacitinib); cytokine targeting (for example, anti-TNF agents such as infliximab, adalimumab, golimumab or certolizumab pegol); and lymphocyte trafficking blockade (anti-α4β7 agents such as vedolizumab, anti-β7 agents such as etrolizumab, anti-mucosal addressin cell adhesion molecule 1 (MADCAM1) agents such as PF-00547659 and sphingosine 1-phosphate (S1P) receptor antagonists such as ozanimod). ILC, innate lymphoid cell.

Digby-Bell et al., 2019. *Nature Reviews Gastroenterology & Hepatology*, 17(1), pp.9-20.

Small molecules: **Oral**

- S1P modulator
- JAK inhibitors, TYK
- Oral integrin inhibitors



Who should be on biologics?

Indications:

- standard treatment not working
- acute severe UC not responding to steroid
- fistulising disease
- extra-intestinal manifestations

  eg pyoderma gangrenosum

Risks of Biologics

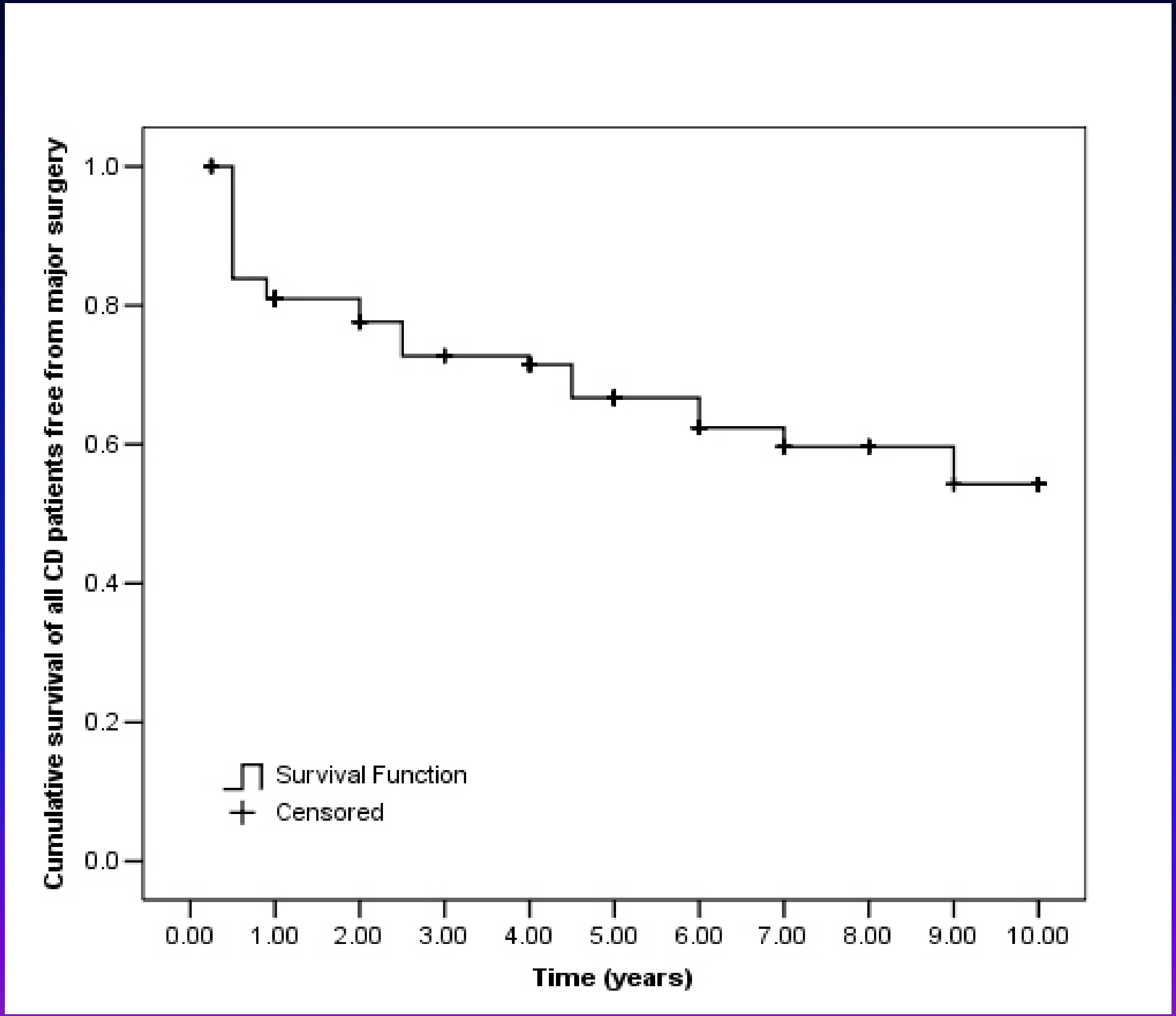
- Infection:
  - reactivation of latent TB & latent viral infection eg HBV (anti-TNFs)
  - Herpes zoster (JAKi)
- Malignancy:
  - Lymphoma (anti-TNFs)

Surgical therapy

- >50% of CD patients require surgery during their life time
- Not curative for CD, disease recurs close to the anastomosis
- Indications:
  - severe bleed, perforation, stricture, fistula, abscess, failed medical treatment, risk of cancer

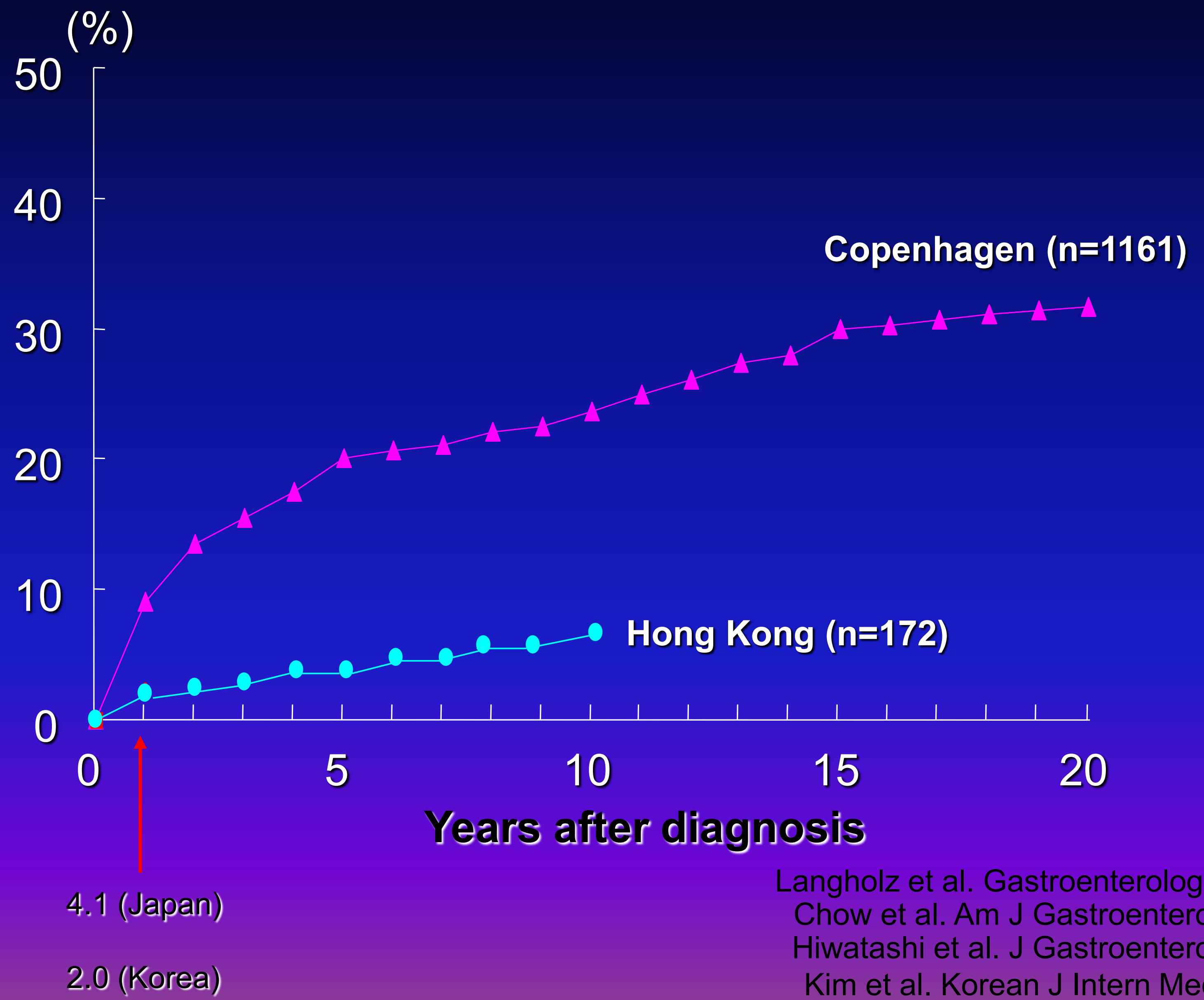
  - Resection of diseased intestine (CD)
  - Stricturoplasty
  - Colectomy, proctocolectomy (acute severe UC and CRC)

Risk of HK CD patients free from major surgery



Chow et al. Inflamm Bowel Dis 2008

Cumulative Colectomy Rate in UC



Langholz et al. Gastroenterology 1992
Chow et al. Am J Gastroenterol 2009
Hiwatashi et al. J Gastroenterol 1995
Kim et al. Korean J Intern Med 2000

Questions?

TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!

Look for today's teacher
and the others you may have missed.

(HKU Portal login required)